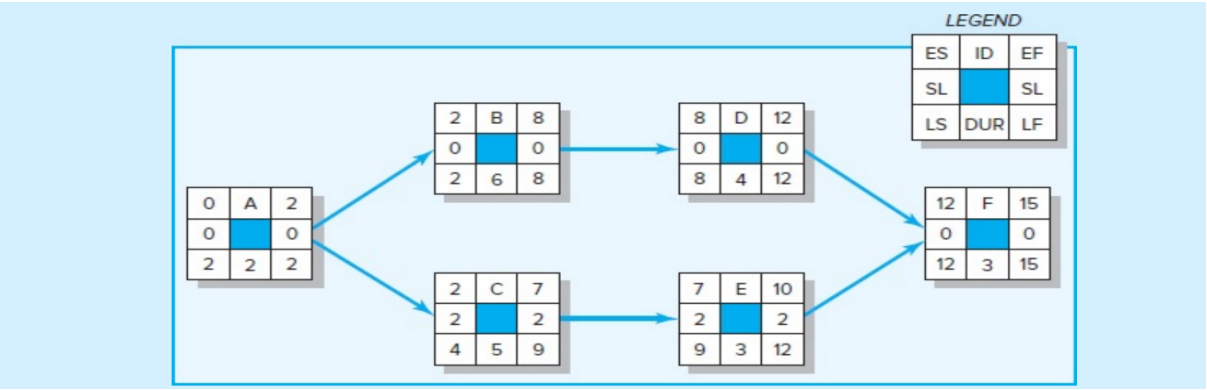
3. Given the following project network and baseline information, complete the form to develop a status report for the project at the end of period 4 and the end of period 8. From the data you have collected and computed for periods 4 and 8, what information are you prepared to tell the customer about the status of the project at the end of period 8?



Project baseline (PV) (in $)

Task	DUR	ES	LF	SL	Budget (PV)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	2	0	2	0	400	200	200														
B	6	2	8	0	2400			200	600	200	600	200	600								
C	5	2	9	2	1500			200	400	500	100	300									
D	4	8	12	0	1600									400	400	400	400				
E	3	7	12	2	900								300	400	200						
F	3	12	15	0	600													200	100	300	
Period PV total						200	200	400	1000	700	700	500	900	800	600	400	400	200	100	300	
Cumulative PV total						200	400	800	1800	2500	3200	3700	4600	5400	6000	6400	6800	7000	7100	7400	

End of Period 4

Task	Actual % Complete	EV	AC	PV	CV	SV
A	Finished	—	300	400	—	—
B	50%	—	1000	800	—	—
C	33%	—	500	600	—	—
D	0%	—	0	—	—	—
E	0%	—	—	—	—	—
Cumulative Totals		—	—	—	—	—

End of Period 8

Task	Actual % Complete	EV	AC	PV	CV	SV
A	Finished	—	300	400	—	—
B	Finished	—	2200	2400	—	—
C	Finished	—	1500	1500	—	—
D	25%	—	300	0	—	—
E	33%	—	300	—	—	—
F	0%	—	0	—	—	—
Cumulative Totals		—	—	—	—	—

Ans:

1. Baseline info (given)

Budgets (BAC) per task:

- A = $400
- B = $2,400
- C = $1,500
- D = $1,600
- E = $900
- F = $600

Total BAC = 400 + 2400 + 1500 + 1600 + 900 + 600 = **$7,400**

From the baseline PV table (the one in your screenshot):

- Cumulative **PV at end of period 4** = **$1,800**
- Cumulative **PV at end of period 8** = **$4,600**

Key Earned Value formulas :

- **EV (Earned Value)** = % complete × Budget
- **PV (Planned Value)** = Budgeted value of work **scheduled** to be done by that time
- **AC (Actual Cost)** = Actual money spent (given)
- **CV (Cost Variance)** = EV − AC
- **SV (Schedule Variance)** = EV − PV
- **CPI (Cost Performance Index)** = EV / AC
- **SPI (Schedule Performance Index)** = EV / PV

2. End of Period 4

Given actual status at t = 4:

- A: Finished, AC = 300, Budget = 400
- B: 50%, AC = 1000, Budget = 2400
- C: 33%, AC = 500, Budget = 1500
- D: 0%, AC = 0
- E: 0%, AC = 0

Step 1: EV per activity

1. **A**
   EV = 100% × 400 = **400**
2. **B**
   EV = 50% × 2400 = 0.5 × 2400 = **1200**
3. **C**
   EV = 33% × 1500 ≈ 0.33 × 1500 ≈ **500** (textbook rounds)
4. **D**
   EV = 0% × 1600 = **0**
5. **E**
   EV = 0% × 900 = **0**

Total EV_4 = 400 + 1200 + 500 + 0 + 0 = **$2,100**

Step 2: PV per activity at period 4 (from baseline table)

- A is planned to be fully done → PV_A_4 = **400**
- B planned 1, 2, 3, 4 time units from t=2–8 → by t=4, planned PV_B_4 = **800**
- C planned partial work → PV_C_4 = **600**
- D, E not planned to have started → $PV_D_4 = 0$, $PV_E_4 = 0$

Check:
PV_4 total = 400 + 800 + 600 = **$1,800** (matches table)

Step 3: AC total at period 4

AC_4 total = 300 (A) + 1000 (B) + 500 (C) + 0 + 0 = **$1,800**

Step 4: Variances per activity

1. **A**
   - CV = EV − AC = 400 − 300 = **+100**
   - SV = EV − PV = 400 − 400 = **0**
2. **B**
   - CV = 1200 − 1000 = **+200**
   - SV = 1200 − 800 = **+400**
3. **C**
   - CV = 500 − 500 = **0**
   - SV = 500 − 600 = **−100**
4. **D, E**
   - CV = 0, SV = 0

Totals at period 4

- **CV_4(total)** = $EV_4 - AC_4$ = 2100 − 1800 = **+300**
- **SV_4(total)** = $EV_4 - PV_4$ = 2100 − 1800 = **+300**

Indexes:

- **CPI_4** = EV_4 / AC_4 = 2100 / 1800 ≈ **1.17**
- **SPI_4** = EV_4 / PV_4 = 2100 / 1800 ≈ **1.17**

Interpretation at period 4

- CPI > 1 → doing more work per dollar than planned (**under budget**).
- SPI > 1 → completed more work than planned by time 4 (**ahead of schedule**).

3. End of Period 8 – detailed calculations

Given actual status at t = 8:

- A: Finished, AC = 300
- B: Finished, AC = 2200
- C: Finished, AC = 1500
- D: 25%, AC = 300
- E: 33%, AC = 300
- F: 0%, AC = 0

Step 1: EV per activity

1. **A**
   - EV = 100% × 400 = **400**
2. **B**
   - EV = 100% × 2400 = **2400**
3. **C**
   - EV = 100% × 1500 = **1500**
4. **D**
   - EV = 25% × 1600 = 0.25 × 1600 = **400**
5. **E**
   - EV = 33% × 900 ≈ 0.33 × 900 ≈ **300**
6. **F**
   - EV = 0% × 600 = **0**

Total EV_8 = 400 + 2400 + 1500 + 400 + 300 + 0 = **$5,000**

Step 2: PV at period 8 (from baseline)

From the baseline table (cumulative row):

- **PV_8 total = $4,600**

Distributed roughly as:

- A: 400 (fully planned)
- B: 2400 (fully planned)
- C: 1500 (fully planned)
- E: 300 planned by t=8
- D, F: 0 planned by t=8

(That matches your PV column entries.)

Step 3: AC total at period 8

AC_8 total = 300 + 2200 + 1500 + 300 + 300 + 0
= 300 + 2200 = 2500
2500 + 1500 = 4000
4000 + 300 = 4300
4300 + 300 = **4600**

Step 4: Variances per activity at t = 8

1. **A**
   - CV = 400 − 300 = **+100**
   - SV = 400 − 400 = **0**
2. **B**
   - CV = 2400 − 2200 = **+200**
   - SV = 2400 − 2400 = **0**
3. **C**
   - CV = 1500 − 1500 = **0**
   - SV = 1500 − 1500 = **0**
4. **D**
   - CV = 400 − 300 = **+100**
   - PV_D_8 = 0 → SV = 400 − 0 = **+400**
     (Started earlier than baseline; that's schedule gain.)

5. **E**
    - CV = 300 − 300 = **0**
    - SV = 300 − 300 = **0**
6. **F**
    - CV = 0, SV = 0

Totals at period 8

- **CV_8(total)** = $EV_8 - AC_8$ = 5000 − 4600 = **+400**
- **SV_8(total)** = $EV_8 - PV_8$ = 5000 − 4600 = **+400**

Indexes:

- **CPI_8** = EV_8 / AC_8 = 5000 / 4600 ≈ **1.09**
- **SPI_8** = EV_8 / PV_8 = 5000 / 4600 ≈ **1.09**

4. What you can tell the customer at end of period 8

Putting it into a clean message:

1. **Cost status**
    - You've earned $5,000 of planned value while spending only $4,600.
    - Positive cost variance: **CV = +$400**, **CPI ≈ 1.09**.
    - Meaning: *for every $1 spent, you're getting about $1.09 worth of planned work* → **under budget**.
2. **Schedule status**
    - You planned to have $4,600 worth of work done; you actually have $5,000 worth done.
    - Positive schedule variance: **SV = +$400**, **SPI ≈ 1.09**.
    - Meaning: *you have completed more work than scheduled by this time* → **ahead of schedule**.
3. **Activity-level insight**
    - A, B, C are fully complete within or below budget.
    - D has started earlier than planned (positive SV), giving schedule buffer on downstream path.

- E has started and matches its plan.
- F is not yet started but also not yet scheduled/required by time 8 (per baseline).

4. **Forward-looking statement (nice to add)**
   - If current cost efficiency continues, an estimate at completion:
     - **EAC ≈ BAC / CPI_8 = 7400 / 1.09 ≈ $6,790** (roughly)
   - So, you can credibly say:
     "We expect to finish the project ahead of schedule and under the original $7,400 budget if current performance trends continue."

Project baseline (PV) (in $)

Task	DUR	ES	LF	SL	Budget (PV)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	2	0	2	0	400	200	200														
B	6	2	8	0	2400			200	600	200	600	200	600								
C	5	2	9	2	1500			200	400	500	100	300									
D	4	8	12	0	1600									400	400	400	400				
E	3	7	12	2	900								300	400	200						
F	3	12	15	0	600													200	100	300	
Period PV total						200	200	400	1000	700	700	500	900	800	600	400	400	200	100	300	
Cumulative PV total						200	400	800	1800	2500	3200	3700	4600	5400	6000	6400	6800	7000	7100	7400	

End of Period 4

Task	Actual % Complete	EV	AC	PV	CV	SV
A	Finished	400	300	400	100	0
B	50%	1200	1000	800	200	400
C	33%	500	500	600	0	-100
D	0%	0	0	0	0	0
E	0%	0	0	0	0	0
Cumulative Totals		2100	1800	1800	300	300

End of Period 8

Task	Actual % Complete	EV	AC	PV	CV	SV
A	Finished	400	300	400	100	0
B	Finished	2400	2200	2400	200	0
C	Finished	1500	1500	1500	0	0
D	25%	400	300	0	100	400
E	33%	300	300	300	0	0
F	0%	0	0	0	0	0
Cumulative Totals		5000	4600	4600	400	400

4. Given the following project network, baseline, and status information, develop status reports for periods 2, 4, 6, 8 and complete the performance indexes table. Calculate the EACf and the VACf. Based on your data, what is your assessment of the current status of the project? At completion?

4. Given the following project network, baseline, and status information, develop status reports for periods 2, 4, 6, 8 and complete the performance indexes table. Calculate the EAC_f and the VAC_f. Based on your data, what is your assessment of the current status of the project? At completion?

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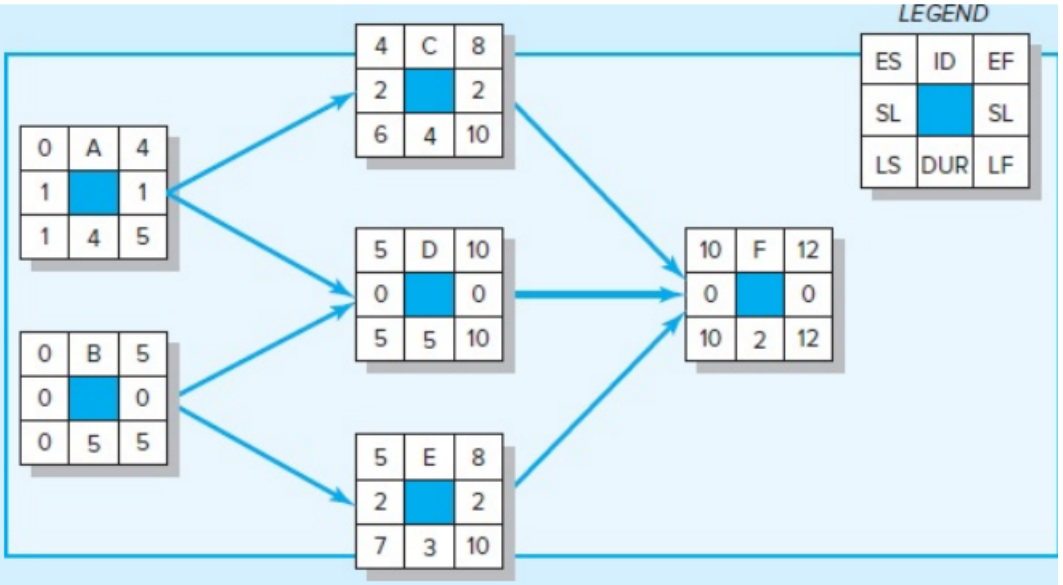


ID	Budget ($000)	0	1	2	3	4	5	6	7	8	9	10	11	12
A	40	10	10	10	10									
B	32	8	4	8	4	8								
C	48					12	12	12	12					
D	18						6	2	2	2	6			
E	28						8	8	12					
F	40											20	20	
Total	206	18	14	18	14	20	26	22	26	2	6	20	20	
Cumulative		18	32	50	64	84	110	132	158	160	166	186	206	

Status Report: Ending Period 2 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	75%	—	25	—	—	—
B	50%	—	12	—	—	—
Cumulative Totals		—	**37**	—	—	—

Status Report: Ending Period 4 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	—	35	—	—	—
B	100%	—	24	—	—	—
Cumulative Totals		—	**59**	—	—	—

Status Report: Ending Period 6 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	—	35	—	—	—
B	100%	—	24	—	—	—
C	75%	—	24	—	—	—
D	0%	—	0	—	—	—
E	50%	—	10	—	—	—
Cumulative Totals		—	**93**	—	—	—

Status Report: Ending Period 8 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	—	35	—	—	—
B	100%	—	24	—	—	—
C	100%	—	32	—	—	—
D	33%	—	20	—	—	—
E	100%	—	20	—	—	—
Cumulative Totals		—	**131**	—	—	—

Performance Indexes Summary

Period	EV	AC	PV	SPI	CPI	PCIB
2	—	—	—	—	—	—
4	—	—	—	—	—	—
6	—	—	—	—	—	—
8	—	—	—	—	—	—

EAC_r = _____ VAC_r = _____

1. Baseline data

Budgets (BAC):

- A = 40
- B = 32
- C = 48
- D = 18

- E = 28
- F = 40

Total BAC = 206

From the baseline PV table (bottom row, “Cumulative”):

- $PV_2 = 32$
- $PV_4 = 64$
- $PV_6 = 110$
- $PV_8 = 158$
- $PV_{12} = 206$

Per-activity planned value pattern (from the same table):

- **A**: 10,10,10,10 over periods 1–4 → PV_A(t): 10,20,30,40.
- **B**: 8,4,8,4,8 over periods 1–5 → PV_B(1–5): 8,12,20,24,32.
- **C**: 12,12,12,12 over periods 5–8 → PV_C(5–8): 12,24,36,48.
- **D**: 6,2,2,2,6 over periods 6–10 → PV_D(6–10): 6,8,10,12,18.
- **E**: 8,8,12 over periods 6–8 → PV_E(6–8): 8,16,28.
- **F**: 20,20 over periods 11–12.

Core formulas:

- **EV** = % complete × BAC
- **CV** = EV − AC
- **SV** = EV − PV
- **CPI** = EV / AC
- **SPI** = EV / PV
- **PCIB** = EV / BAC(total) (Percent Complete Index – Budget)

2. Status Report – End of Period 2

Given:

- A: 75% complete, AC = 25
- B: 50% complete, AC = 12

Task A

- EV = 0.75 × 40 = **30**
- PV at t=2 = 10 + 10 = **20**
- CV = 30 − 25 = **+5**
- SV = 30 − 20 = **+10**

Task B

- EV = 0.50 × 32 = **16**
- PV at t=2 = 8 + 4 = **12**
- CV = 16 − 12 = **+4**
- SV = 16 − 12 = **+4**

Cumulative (t=2)

- EV = 30 + 16 = **46**
- AC = 25 + 12 = **37**
- PV = 20 + 12 = **32**
- CV = 46 − 37 = **+9**
- SV = 46 − 32 = **+14**

Indexes:

- SPI_2 = 46 / 32 = **1.44**
- CPI_2 = 46 / 37 ≈ **1.24**
- $PCIB_2$ = 46 / 206 ≈ **0.223** → **22.3%** of total budgeted work earned

3. Status Report – End of Period 4

Given:

- A: 100%, AC = 35
- B: 100%, AC = 24

Task A

- EV = 1.00 × 40 = **40**
- $PV_4(A)$ = **40**
- CV = 40 − 35 = **+5**
- SV = 40 − 40 = **0**

Task B

- EV = 1.00 × 32 = **32**
- $PV_4(B)$ = 8 + 4 + 8 + 4 = **24**
- CV = 32 − 24 = **+8**
- SV = 32 − 24 = **+8**

Cumulative (t=4)

- EV = 40 + 32 = **72**
- AC = 35 + 24 = **59**
- PV = 40 + 24 = **64**
- CV = 72 − 59 = **+13**
- SV = 72 − 64 = **+8**

Indexes:

- SPI_4 = 72 / 64 = **1.13**
- CPI_4 = 72 / 59 ≈ **1.22**
- $PCIB_4$ = 72 / 206 ≈ **0.350** → **35.0%**

4. Status Report – End of Period 6

Given:

- A: 100%, AC = 35
- B: 100%, AC = 24
- C: 75%, AC = 24
- D: 0%, AC = 0
- E: 50%, AC = 10

Task A

- EV = 40
- $PV_6(A) = 40$
- CV = 40 − 35 = **+5**
- SV = 40 − 40 = **0**

Task B

- EV = 32
- $PV_6(B) = 32$ (all of B planned by period 5)
- CV = 32 − 24 = **+8**
- SV = 32 − 32 = **0**

Task C

- EV = 0.75 × 48 = **36**
- $PV_6(C) = 12 + 12 =$ **24**
- CV = 36 − 24 = **+12**
- SV = 36 − 24 = **+12**

Task D

- EV = 0 × 18 = **0**
- $PV_6(D) =$ **6**
- CV = 0 − 0 = **0**
- SV = 0 − 6 = **−6**

Task E

- EV = 0.5 × 28 = **14**
- $PV_6(E)$ = **8**
- CV = 14 − 10 = **+4**
- SV = 14 − 8 = **+6**

Cumulative (t=6)

- EV = 40 + 32 + 36 + 0 + 14 = **122**
- AC = 35 + 24 + 24 + 0 + 10 = **93**
- PV = 40 + 32 + 24 + 6 + 8 = **110**
- CV = 122 − 93 = **+29**
- SV = 122 − 110 = **+12**

Indexes:

- SPI_6 = 122 / 110 ≈ **1.11**
- CPI_6 = 122 / 93 ≈ **1.31**
- $PCIB_6$ = 122 / 206 ≈ **0.592** → **59.2%**

5. Status Report – End of Period 8

Given:

- A: 100%, AC = 35
- B: 100%, AC = 24
- C: 100%, AC = 32
- D: 33%, AC = 20
- E: 100%, AC = 20

(Use 33% of D → 1/3 of 18 = **6**)

Task A

- EV = 40
- $PV_8(A)$ = 40
- CV = 40 − 35 = **+5**

- SV = 40 − 40 = **0**

Task B

- EV = 32
- $PV_8(B) = 32$
- CV = 32 − 24 = **+8**
- SV = 32 − 32 = **0**

Task C

- EV = 48
- $PV_8(C) = 48$
- CV = 48 − 32 = **+16**
- SV = 48 − 48 = **0**

Task D

- EV = 0.33 × 18 ≈ **6**
- $PV_8(D) = 6 + 2 + 2 =$ **10**
- CV = 6 − 20 = **−14**
- SV = 6 − 10 = **−4**

Task E

- EV = 28
- $PV_8(E) = 28$
- CV = 28 − 20 = **+8**
- SV = 28 − 28 = **0**

Cumulative (t=8)

- EV = 40 + 32 + 48 + 6 + 28 = **154**
- AC = 35 + 24 + 32 + 20 + 20 = **131**
- PV = 40 + 32 + 48 + 10 + 28 = **158**
- CV = 154 − 131 = **+23**
- SV = 154 − 158 = **−4**

Indexes:

- SPI_8 = 154 / 158 ≈ **0.97**
- CPI_8 = 154 / 131 ≈ **1.18**
- $PCIB_8$ = 154 / 206 ≈ **0.748 → 74.8%**

6. Performance Indexes Summary Table

Period	EV	AC	PV	SPI	CPI	PCIB
2	46	37	32	1.44	1.24	0.22
4	72	59	64	1.13	1.22	0.35
6	122	93	110	1.11	1.31	0.59
8	154	131	158	0.97	1.18	0.75

(PCIB is fraction of total BAC earned.)

6. $EAC_{(f)}$ and $VAC_{(f)}$

We're asked for **$EAC_{(f)}$** and **$VAC_{(f)}$**, using cumulative **CPI at period 8**:

- BAC = 206
- CPI_8 ≈ 154 / 131 ≈ **1.18**

$EAC_{(f)}$ (forecasted cost at completion using CPI):

$$EAC_f = \frac{BAC}{CPI_8} = \frac{206}{154/131} \approx 175.2$$

$VAC_{(f)}$ (variance at completion forecast):

$$VAC_f = BAC - EAC_f \approx 206 - 175.2 \approx +30.8$$

So, the project is forecast to finish about **$30.8K under budget**.

7. Assessment

Current status (end of period 8)

- Cost: **Strongly under budget** (CPI_8 ≈ 1.18, CV = +23).
- Schedule: **Slightly behind plan** (SPI_8 ≈ 0.97, SV = −4).
  The slippage is driven by activity D (started later / progressing slower than planned).

At completion (based on CPI trend)

- Expected final cost ≈ **175K vs BAC 206K** → healthy cost underrun.
- With SPI close to 1 and most major tasks complete, the project should still be able to finish roughly on time if D and F are controlled, while remaining under budget.

$EAC_{(f)}$ and $VAC_{(f)}$

Using CPI at period 8:

$$EAC_f = \frac{BAC}{CPI_8} = \frac{206}{154/131} \approx 175.2$$

$$VAC_f = BAC - EAC_f \approx 206 - 175.2 = +30.8$$

So, in this format:

- **$EAC_{(f)}$ = 175 (≈ $175,000)**
- **$VAC_{(f)}$ = +31 (≈ $31,000 under budget)**

Status Report: Ending Period 2 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	75%	30	25	20	5	10
B	50%	16	12	12	4	4
Cumulative Totals		46	37	32	9	14

Status Report: Ending Period 4 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	24	8	8
Cumulative Totals		72	59	64	13	8

Status Report: Ending Period 6 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	32	8	0
C	75%	36	24	24	12	12
D	0%	0	0	6	0	-6
E	50%	14	10	8	4	6
Cumulative Totals		122	93	110	29	12

Status Report: Ending Period 8 ($000)

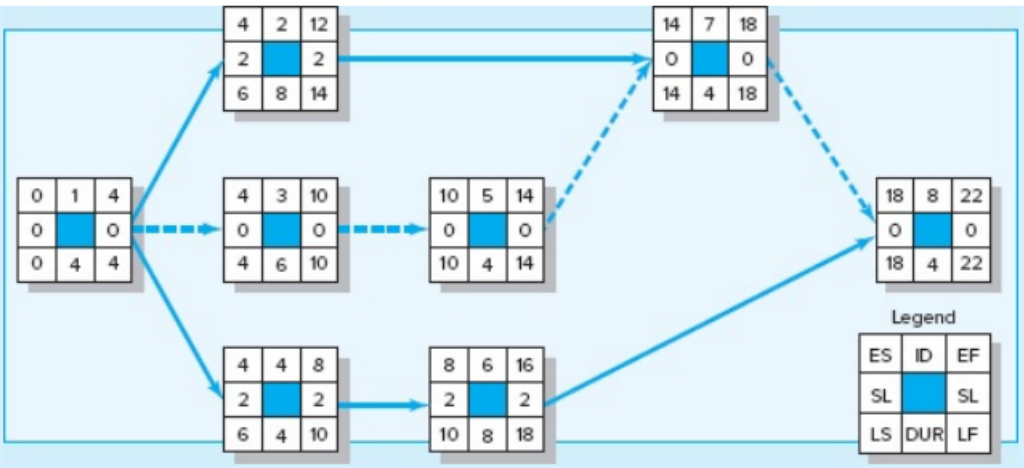
Task	% Complete	EV	AC	PV	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	32	8	0
C	100%	48	32	48	16	0
D	33%	6	20	10	-14	-4
E	100%	28	20	28	8	0
Cumulative Totals		154	131	158	23	4

Performance Indexes Summary

Period	EV	AC	PV	SPI	CPI	PCIB
2	46	37	32	1.44	1.24	0.22
4	72	59	64	1.13	1.22	0.35
6	122	93	110	1.11	1.31	0.59
8	154	131	158	0.97	1.18	0.73

EAC_f = 175 VAC_f = 31

7. The following data have been collected for a British healthcare IT project for two-week reporting periods 2 through 12. Compute the SV, CV, SPI, and CPI for each period. Plot the EV and the AC on the summary graph provided. Plot the SPI, CPI, and PCIB on the index graph provided. (You may use your own graphs.) What is your assessment of the project at the end of period 12?



Task	DUR	ES	LF	SL	PV ($00)	Baseline (PV) ($00) 0–2	2–4	4–6	6–8	8–10	10–12	12–14	14–16	16–18	18–20	20–22
1	4	0	4	0	8	4	4									
2	8	4	14	2	40			10	10	10	10					
3	6	4	10	0	30			10	15	5						
4	4	4	10	2	20			10	10							
5	4	10	14	0	40						20	20				
6	8	8	18	2	60					20	20	10	10			
7	4	14	18	0	20								10	10		
8	4	18	22	0	30										20	10
Period PV total						4	4	30	35	35	50	30	20	10	20	10
Cumulative PV total						4	8	38	73	108	158	188	208	218	238	248

Status Report: Ending Period 2 **($00)**

Task	%Complete	EV	AC	PV	CV	SV
1	50%	—	4	—	—	—
Cumulative Totals		—	**4**	—	—	—

Status Report: Ending Period 4 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	—	10	—	—	—
Cumulative Totals		—	**10**	—	—	—

Status Report: Ending Period 6 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	—	10	—	—	—
2	25%	—	15	—	—	—
3	33%	—	12	—	—	—
4	0%	—	0	—	—	—
Cumulative Totals		—	**37**	—	—	—

Status Report: Ending Period 8 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	—	10	—	—	—
2	30%	—	20	—	—	—
3	60%	—	25	—	—	—
4	0%	—	0	—	—	—
Cumulative Totals		—	**55**	—	—	—

Status Report: Ending Period 10 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	—	10	—	—	—
2	60%	—	30	—	—	—
3	Finished	—	40	—	—	—
4	50%	—	20	—	—	—
5	0%	—	0	—	—	—
6	30%	—	24	—	—	—
Cumulative Totals		—	**124**	—	—	—

Status Report: Ending Period 12 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	—	10	—	—	—
2	Finished	—	50	—	—	—
3	Finished	—	40	—	—	—
4	Finished	—	40	—	—	—
5	50%	—	30	—	—	—
6	50%		40	—	—	—
Cumulative Totals		—	**210**	—	—	—

Period	SPI	CPI	PCIB
2	—	—	—
4	—	—	—
6	—	—	—
8	—	—	—
10	—	—	—
12	—	—	—

SPI = EV/PV
CPI = EV/AC
PCIB = EV/BAC

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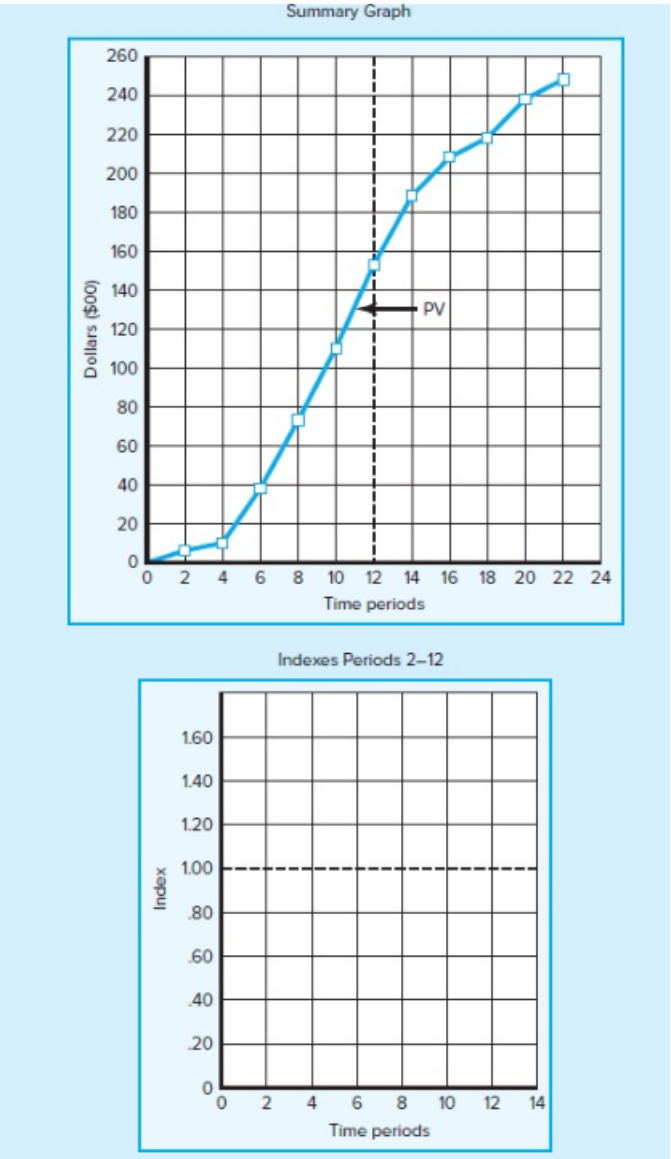


Ans:

1. Baseline recap

Budgets (from PV column):

- Task 1 = 8
- Task 2 = 40
- Task 3 = 30
- Task 4 = 20
- Task 5 = 40

- Task 6 = 60
- Task 7 = 20
- Task 8 = 30

BAC = 8+40+30+20+40+60+20+30 = 248

Cumulative **PV** from the baseline row:

Period	2	4	6	8	10	12	14	16	18	20	22
PVcum	4	8	38	73	108	158	188	208	218	238	248

Up to period 12, tasks 5–8 have **no planned value yet**.

Formulas we use:

- **EV** = (% complete) × BAC(task)
- **CV** = EV − AC
- **SV** = EV − PV
- **SPI** = EV / PV
- **CPI** = EV / AC
- **PCIB** = EV / BAC(total)

For 33% I'll use 1/3 exactly (so 0.33 of 30 = 10).

2. Status Report – Ending Period 2

Given: Task 1 = 50%, AC = 4.

Task 1

- $EV_1 = 0.5 \times 8 =$ **4**
- PV_1 at t=2 = **4**
- $CV_1 = 4 - 4 =$ **0**
- $SV_1 = 4 - 4 =$ **0**

Cumulative

- EV = 4, AC = 4, PV = 4, CV = 0, SV = 0

3. Status Report – Ending Period 4

Given: Task 1 finished, AC = 10.

Task 1

- $EV_1 = 1.0 \times 8 =$ **8**
- PV_1 at t=4 = **8**
- $CV_1 = 8 - 10 =$ **−2**
- $SV_1 = 8 - 8 =$ **0**

Cumulative

- EV = 8, AC = 10, PV = 8, CV = −2, SV = 0

4. Status Report – Ending Period 6

Given:
1 finished (AC 10)
2 at 25% (AC 15)
3 at 33% (AC 12)
4 at 0% (AC 0)

Task 1

- $EV_1 = 8$; $PV_1(6) = 8 \rightarrow CV_1 = -2$; $SV_1 = 0$

Task 2

- $EV_2 = 0.25 \times 40 =$ **10**
- $PV_2(6) = 10$
- $CV_2 = 10 - 15 =$ **−5**
- $SV_2 = 10 - 10 =$ **0**

Task 3

- $EV_3 = (1/3) \times 30 =$ **10**
- $PV_3(6) = 10$
- $CV_3 = 10 - 12 =$ **−2**
- $SV_3 = 10 - 10 =$ **0**

Task 4

- $EV_4 = 0$; $PV_4(6) = 10$
- $CV_4 = 0 - 0 = 0$
- $SV_4 = 0 - 10 =$ **−10**

Cumulative

- EV = 8 + 10 + 10 + 0 = **28**
- AC = 10 + 15 + 12 + 0 = **37**
- PV = 8 + 10 + 10 + 10 = **38**
- CV = 28 − 37 = **−9**
- SV = 28 − 38 = **−10**

5. Status Report – Ending Period 8

Given:
1 finished (AC 10)
2 at 30% (AC 20)
3 at 60% (AC 25)
4 at 0% (AC 0)

Task 1

- $EV_1 = 8$; $PV_1(8)=8 \rightarrow CV_1 = -2$; $SV_1 = 0$

Task 2

- $EV_2 = 0.30 \times 40 =$ **12**
- $PV_2(8) = 20$
- $CV_2 = 12 - 20 =$ **−8**
- $SV_2 = 12 - 20 =$ **−8**

Task 3

- $EV_3 = 0.60 \times 30 =$ **18**
- $PV_3(8) = 25$
- $CV_3 = 18 - 25 =$ **−7**
- $SV_3 = 18 - 25 =$ **−7**

Task 4

- $EV_4 = 0$; $PV_4(8) = 20$
- $CV_4 = 0 - 0 = 0$
- $SV_4 = 0 - 20 =$ **−20**

Cumulative

- EV = 8 + 12 + 18 + 0 = **38**
- AC = 10 + 20 + 25 + 0 = **55**
- PV = 8 + 20 + 25 + 20 = **73**
- CV = 38 − 55 = **−17**
- SV = 38 − 73 = **−35**

6. Status Report – Ending Period 10

Given:
1 finished (AC 10)
2 at 60% (AC 30)
3 finished (AC 40)
4 at 50% (AC 20)
5 at 0% (AC 0)
6 at 30% (AC 24)

Task 1

- $EV_1 = 8$; $PV_1(10)=8 \rightarrow CV_1 = -2$; $SV_1 = 0$

Task 2

- $EV_2 = 0.60 \times 40 =$ **24**
- $PV_2(10) = 30$
- $CV_2 = 24 - 30 =$ **−6**
- $SV_2 = 24 - 30 =$ **−6**

Task 3

- $EV_3 = 1.0 \times 30 =$ **30**
- $PV_3(10) = 30$
- $CV_3 = 30 - 40 =$ **−10**
- $SV_3 = 30 - 30 =$ **0**

Task 4

- $EV_4 = 0.50 \times 20 =$ **10**
- $PV_4(10) = 20$
- $CV_4 = 10 - 20 =$ **−10**
- $SV_4 = 10 - 20 =$ **−10**

Task 5

- $EV_5 = 0$, $PV_5(10) = 0 \rightarrow CV_5 = 0$, $SV_5 = 0$

Task 6

- $EV_6 = 0.30 \times 60 =$ **18**
- $PV_6(10) = 0$ (starts later)
- $CV_6 = 18 - 24 =$ **−6**
- $SV_6 = 18 - 0 =$ **+18**

Cumulative

- EV = 8 + 24 + 30 + 10 + 0 + 18 = **90**
- AC = 10 + 30 + 40 + 20 + 0 + 24 = **124**
- PV = 108
- CV = 90 − 124 = **−34**
- SV = 90 − 108 = **−18**

7. Status Report – Ending Period 12

Given:
1 finished (AC 10)
2 finished (AC 50)
3 finished (AC 40)
4 finished (AC 40)
5 at 50% (AC 30)
6 at 50% (AC 40)

Task 1

- $EV_1 = 8$; $PV_1(12)=8 \rightarrow CV_1 = -2$; $SV_1 = 0$

Task 2

- $EV_2 = 40$; $PV_2(12)=40 \rightarrow CV_2 = 40 - 50 =$ **−10** ; $SV_2 = 0$

Task 3

- $EV_3 = 30$; $PV_3(12)=30 \rightarrow CV_3 = 30 - 40 =$ **−10** ; $SV_3 = 0$

Task 4

- $EV_4 = 20$; $PV_4(12)=20 \rightarrow CV_4 = 20 - 40 =$ **−20** ; $SV_4 = 0$

Task 5

- $EV_5 = 0.5 \times 40 =$ **20**
- $PV_5(12) = 0$ (starts at 14)
- $CV_5 = 20 - 30 =$ **−10**
- $SV_5 = 20 - 0 =$ **+20**

Task 6

- $EV_6 = 0.5 \times 60 =$ **30**
- $PV_6(12) = 0$
- $CV_6 = 30 - 40 =$ **−10**
- $SV_6 = 30 - 0 =$ **+30**

Cumulative

- EV = 8 + 40 + 30 + 20 + 20 + 30 = **148**
- AC = 10 + 50 + 40 + 40 + 30 + 40 = **210**
- PV = 158
- CV = 148 − 210 = **−62**
- SV = 148 − 158 = **−10**

8. Performance Indexes (using cumulative values)

For each status period:

1. **Period 2**
   SPI = 4 / 4 = **1.00**
   CPI = 4 / 4 = **1.00**
   PCIB = 4 / 248 ≈ **0.02**
2. **Period 4**
   SPI = 8 / 8 = **1.00**
   CPI = 8 / 10 = **0.80**
   PCIB = 8 / 248 ≈ **0.03**

3. **Period 6**
   SPI = 28 / 38 ≈ **0.74**
   CPI = 28 / 37 ≈ **0.76**
   PCIB = 28 / 248 ≈ **0.11**

4. **Period 8**
   SPI = 38 / 73 ≈ **0.52**
   CPI = 38 / 55 ≈ **0.69**
   PCIB = 38 / 248 ≈ **0.15**

5. **Period 10**
   SPI = 90 / 108 ≈ **0.83**
   CPI = 90 / 124 ≈ **0.73**
   PCIB = 90 / 248 ≈ **0.36**

6. **Period 12**
   SPI = 148 / 158 ≈ **0.94**
   CPI = 148 / 210 ≈ **0.70**
   PCIB = 148 / 248 ≈ **0.60**

9. Forecast: $EAC_{(f)}$ and $VAC_{(f)}$

Use cumulative CPI at period 12:

- CPI_{12} = 148 / 210 ≈ **0.705**

$EAC_{(f)}$ (forecast cost if current cost efficiency continues):

$$EAC_f = \frac{BAC}{CPI_{12}} = \frac{248}{148/210} = \frac{248 \times 210}{148} \approx 352$$

$VAC_{(f)}$ (variance at completion):

$$VAC_f = BAC - EAC_f \approx 248 - 352 = -104$$

So, the project is trending about **$10,400 over budget**.

$EAC_{(f)}$ and $VAC_{(f)}$ (using CPI at period 12)

- **CPI_{12} = EV/AC = 148 / 210 ≈ 0.705**
- **$EAC_{(f)}$ = BAC / CPI_{12} = 248 / 0.705 ≈ 352**
- **$VAC_{(f)}$ = BAC – $EAC_{(f)}$ ≈ 248 – 352 = -104**

Status Report: Ending Period 2 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	50%	4	4	4	0	0
Cumulative Totals		4	4	4	0	0

Status Report: Ending Period 4 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
Cumulative Totals		8	10	8	-2	0

Status Report: Ending Period 6 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	25%	10	15	10	-5	0
3	33%	10	12	10	-2	0
4	0%	0	0	10	0	-10
Cumulative Totals		28	37	38	-9	-10

Status Report: Ending Period 8 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	30%	12	20	20	-8	-8
3	60%	18	25	25	-7	-7
4	0%	0	0	20	0	-20
Cumulative Totals		38	55	73	-17	-35

Status Report: Ending Period 10 ($00)

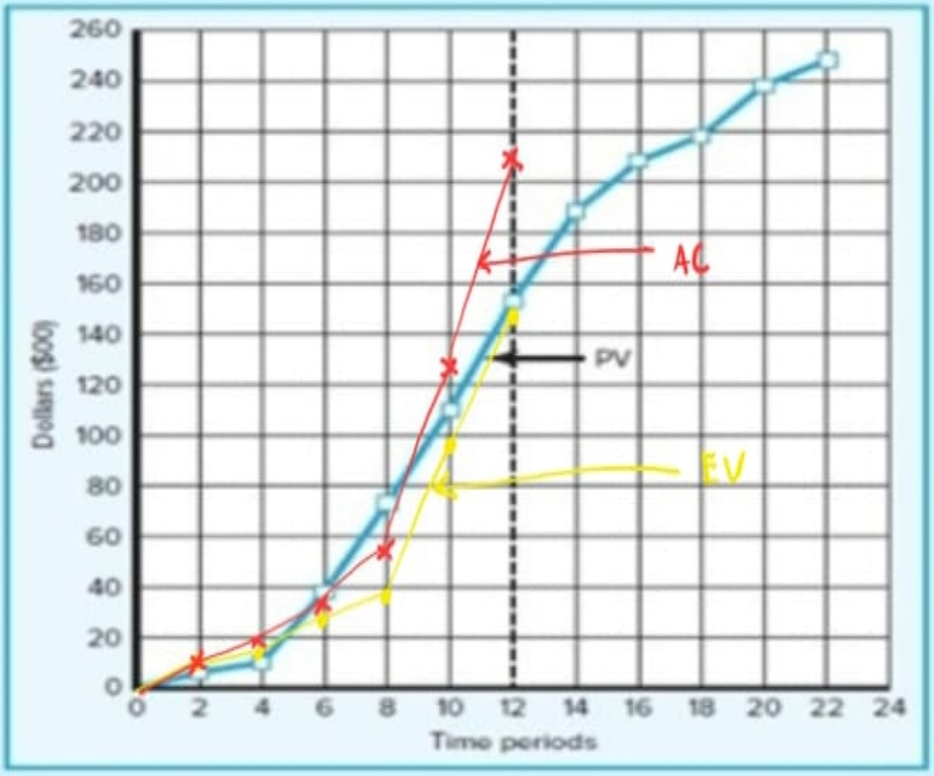
Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	60%	24	30	30	-6	-6
3	Finished	30	40	30	-10	0
4	50%	10	20	20	-10	-10
5	0%	0	0	0	0	0
6	30%	18	24	20	-6	-2
Cumulative Totals		90	124	108	-34	-18

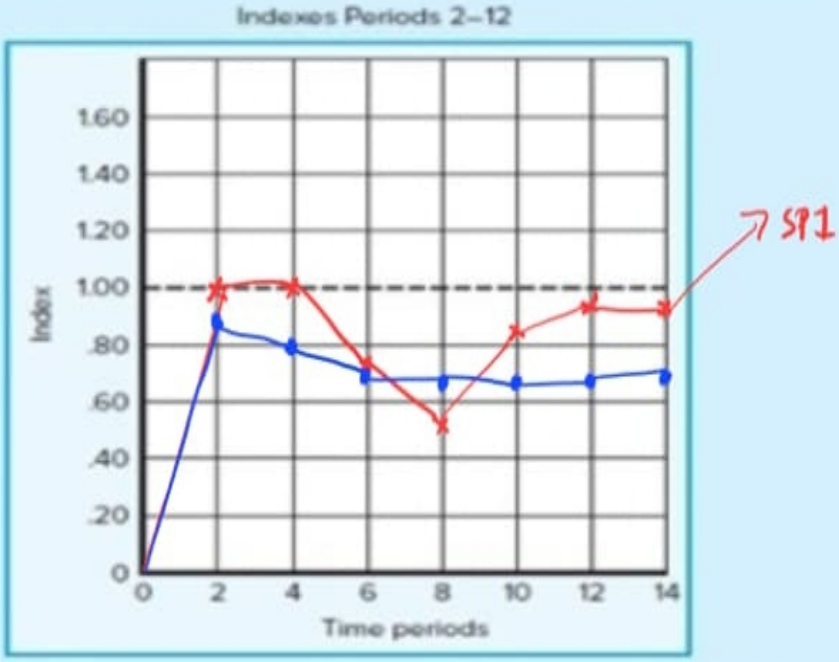
Status Report: Ending Period 12 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	Finished	40	50	40	-10	0
3	Finished	30	40	30	-10	0
4	Finished	20	40	20	-20	0
5	50%	20	30	20	-10	0
6	50%	30	40	40	-10	-10
Cumulative Totals		148	210	158	-62	-10

Period	SPI	CPI	PCIB
2	1	1	0.02
4	1	0.8	0.03
6	0.74	0.76	0.11
8	0.52	0.69	0.15
10	0.83	0.73	0.36
12	0.74	0.7	0.60

SPI = EV/PV
CPI = EV/AC
PCIB = EV/BAC


Summary Graph
Dollars ($00)
Time periods
AC
PV
EV


Indexes Periods 2–12
Index
Time periods
SPI

13.3 Scanner Project

You have been serving as Electroscan's project manager and are now well along in the project. Develop a narrative status report for the board of directors of the chain store that discusses the status of the project to date (see Table C13.3) and at completion. Be as specific as you can, using the numbers given and those you might develop. Remember, your audience is not familiar with the jargon used by project managers and computer software personnel; therefore, some explanation may be necessary. Your report will be evaluated on your detailed use of the data, your total perspective of the current status and future status of the project, and your recommended changes (if any).

Case 13.3

Scanner Project

You have been serving as Electroscan's project manager and are now well along in the project. Develop a narrative status report for the board of directors of the chain store that discusses the status of the project to date (see Table C13.3) and at completion. Be as specific as you can, using the numbers given and those you might develop. Remember, your audience is not familiar with the jargon used by project managers and computer software personnel; therefore, some explanation may be necessary. Your report will be evaluated on your detailed use of the data, your total perspective of the current status and future status of the project, and your recommended changes (if any).

TABLE C13.3

Electroscan, Inc.
555 Acorn Street, Suite 5
Boston, Massachusetts

29 In-store Scanner Project
(thousands of dollars)
Actual Progress as of January 1

Name	PV	EV	AC	SV	CV	BAC	EAC_f
Scanner project	420	395	476	−25	−81	915	1103
H 1.0 Hardware	92	88	72	−4	16	260	213
H 1.1 Hardware specifications (DS)	20	20	15	0	5	20	15
H 1.2 Hardware design (DS)	30	30	25	0	5	30	25
H 1.3 Hardware documentation (DOC)	10	6	5	−4	1	10	8
H 1.4 Prototypes (PD)	2	2	2	0	0	40	40
H 1.5 Test prototypes (T)	0	0	0	0	0	30	30
H 1.6 Order circuit boards (PD)	30	30	25	0	5	30	25
H 1.7 Preproduction models (PD)	0	0	0	0	0	100	100
OP 1.0 Operating system	195	150	196	−45	−46	330	431
OP 1.1 Kernel specifications (DS)	20	20	15	0	5	20	15
OP 1.2 Drivers	45	55	76	10	−21	70	97
OP 1.2.1 Disk drivers (DEV)	25	30	45	5	−15	40	60
OP 1.2.2 I/O drivers (DEV)	20	25	31	5	−6	30	37
OP 1.3 Code software	130	75	105	−55	−30	240	336
OP 1.3.1 Code software (C)	30	20	40	−10	−20	100	200
OP 1.3.2 Document software (DOC)	45	30	25	−15	5	50	42
OP 1.3.3 Code interfaces (C)	55	25	40	−30	−15	60	96
OP 1.3.4 Beta test software (T)	0	0	0	0	0	30	30
U 1.0 Utilities	87	108	148	21	−40	200	274
U 1.1 Utilities specifications (DS)	20	20	15	0	5	20	15
U 1.2 Routine utilities (DEV)	20	20	35	0	−15	20	35
U 1.3 Complex utilities (DEV)	30	60	90	30	−30	100	150
U 1.4 Utilities documentation (DOC)	17	8	8	−9	0	20	20
U 1.5 Beta test utilities (T)	0	0	0	0	0	40	40
S 1.0 System integration	46	49	60	3	−11	125	153
S 1.1 Architecture decisions (DS)	9	9	7	0	2	10	8
S 1.2 Integration hard/soft (DEV)	25	30	45	5	−15	50	75
S 1.3 System hard/software test (T)	0	0	0	0	0	20	20
S 1.4 Project documentation (DOC)	12	10	8	−2	2	15	12
S 1.5 Integration acceptance testing (T)	0	0	0	0	0	30	30

Ans:

1. Big picture – where we are today

From the "Scanner project" summary row:

- **Planned Value (PV)** = 420
  → What we *expected* to have completed by now.
- **Earned Value (EV)** = 395
  → Budgeted value of the work actually *finished*.
- **Actual Cost (AC)** = 476
  → What we have actually *spent* so far.

Key variances

- **Schedule Variance (SV)** = EV − PV = 395 − 420 = **−25**
  → We have completed $25K less work than planned (slightly **behind schedule**).
- **Cost Variance (CV)** = EV − AC = 395 − 476 = **−81**
  → We have spent $81K more than the value of the work accomplished (**over budget**).

Performance indexes

- **Schedule Performance Index (SPI)** = EV / PV = 395 / 420 ≈ **0.94**

We are progressing at about **94% of the planned rate**.

- **Cost Performance Index (CPI)** = EV / AC = 395 / 476 ≈ **0.83**

For every $1.00 spent, we're only getting about **$0.83** of planned value.
This is the central concern.

Forecast at completion

- **Original total budget (BAC)** = 915
- Using current CPI as our efficiency estimate:

$$EAC_f = \frac{BAC}{CPI} = \frac{915}{0.83} \approx 1{,}103$$

- **Forecast overrun ($VAC_{(f)}$)** = BAC − EAC_f ≈ 915 − 1,103 = **−188**

In plain terms:
If we continue the way we're performing now, the Scanner Project is expected to finish about **$188K (≈ 21%) over budget**, with a modest schedule delay risk.

2. How each major area is performing

Summaries use the same EV logic but rolled up by major deliverable (from Table C13.3).

2.1 Hardware (H 1.0)

- PV = 92, EV = 88, AC = 72
- **SPI ≈ 0.96** (slightly behind)
- **CPI ≈ 1.22** (spending less than planned)
- $EAC_{(f)}$ given = 213 vs BAC 260 → **likely to underrun budget**

Message: Hardware is under good control: essentially on schedule and **saving money**. No major corrective action needed.

2.2 Operating System (OP 10)

- PV = 195, EV = 150, AC = 196
- **SV = −45** → substantially behind.
- **CV = −46** → over budget.
- **SPI ≈ 0.77**, **CPI ≈ 0.77**
- $EAC_{(f)}$ = 431 vs BAC 330 → **projected overrun ≈ $101K**

Message (non-jargon):
This is one of our core problem areas. Progress is slow *and* expensive. At current productivity, OS work alone could exceed its budget by about a third.

2.3 Utilities (U 10)

- PV = 87, EV = 108, AC = 148
- **SV = +21** → ahead of schedule.
- **CV = −40** → over budget.
- **SPI ≈ 1.24**, **CPI ≈ 0.73**
- $EAC_{(f)}$ = 274 vs BAC 200 → **projected overrun ≈ $74K**

Message:
The utilities team is moving fast but burning cash heavily. They're getting work done earlier than planned, but at a significantly higher cost per unit of output.

2.4 System Integration (S 10)

- PV = 46, EV = 49, AC = 60
- **SV = +3** → slightly ahead.
- **CV = −11** → over budget.
- **SPI ≈ 1.07**, **CPI ≈ 0.82**
- $EAC_{(f)}$ = 153 vs BAC 125 → **projected overrun ≈ $28K**

Message:
Integration is on (or a bit ahead of) schedule, but also trending over budget—likely due to early troubleshooting and coordination effort.

3. What this means for the board (in plain language)

1. **Overall status**
    - We are **slightly behind schedule** (about 6% less work done than planned).
    - We are **seriously over-spending** (about 17% cost overrun so far).
    - If performance does not improve, total cost could reach around **$1.1M vs the planned $915K**.
2. **Where the trouble is**
    - **Operating System development**: behind and overspending → largest single driver of risk.
    - **Utilities and System Integration**: technically making progress, but at a poor cost efficiency.
    - **Hardware**: a bright spot; good model of cost control and planning discipline.
3. **Risk to delivery**
    - If OS and integration issues persist, we risk:
        - Delays in stable software availability,
        - Increased integration rework,
        - Further cost growth beyond the current forecast.

4. Recommended actions

These are framed so a non-technical board understands exactly what we'll do.

4.1 Stabilize the Operating System work

- **Immediate root-cause review** within two weeks:
    - Are we facing underestimated complexity?
    - Too many change requests?
    - Skill/resource mismatch?
- **Actions likely needed:**
    - Lock down scope and freeze non-critical changes.
    - Reassign senior engineers to critical OS modules.
    - Introduce short, time-boxed milestones with clear acceptance tests.
- **Board message:** this area needs focused intervention; we will return with an updated forecast after the review.

4.2 Contain Utilities and Integration costs

- For **Utilities**:
    - Clarify why effort is so high (gold plating? redundant work?).
    - Enforce tighter time reporting and approval for overtime / extra features.
- For **System Integration**:
    - Ensure integration is not compensating for unstable OS/Utilities by "patching around" problems.
    - Tie further integration work to stable component baselines.

4.3 Leverage Hardware performance

- Keep Hardware team's practices as the standard:
    - Clear scope, realistic estimates, disciplined change control.
- Use their lead as buffer for later system testing, not as an excuse to relax controls elsewhere.

4.4 Update forecast and decision points

- Maintain **monthly EV reviews**:
    - Track CPI & SPI by major deliverable.
    - If combined CPI does not improve toward **0.9+** in the next two periods, present the board with explicit options:
        - Approve a revised budget and schedule, **or**
        - De-scope or phase features to protect total investment.

5. Concise narrative you can hand in / present

As of January 1, the Scanner Project has earned $395K of planned work against a plan of $420K and has spent $476K. This places us slightly behind schedule (SV = −$25K; SPI = 0.94) and significantly over budget (CV = −$81K; CPI = 0.83). If current performance continues, we forecast a final cost of approximately $1.10M compared with the approved $915K budget, an overrun of about $188K.

Hardware development is essentially on schedule and under its allocated budget, providing confidence in that portion of the design. However, operating system development is both behind schedule and over budget and, together with the Utilities and System Integration work packages, is driving most of the current and projected overrun. Utilities work is ahead of schedule but at an unacceptably high cost, while integration is slightly ahead of schedule but trending over budget.

Our recommendation is to immediately target the operating system and utilities work for corrective action: freeze nonessential changes, verify requirements, realign staffing, and

implement short, measurable milestones. We will maintain monthly earned value reviews and report back revised forecasts. Without corrective action, we expect the project to deliver the intended functionality, but at a materially higher cost; with focused intervention now, we believe we can reduce the overrun and protect both schedule and business objectives.